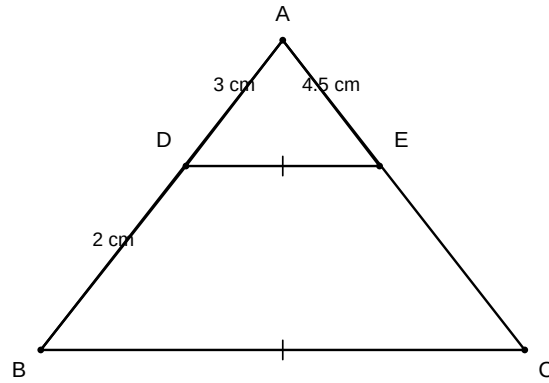


Diagram Test — Solutions

1. 1. In $\triangle ABC$, D and E lie on AB and AC respectively such that $DE \parallel BC$. If $AD = 3$ cm, $DB = 2$ cm and $AE = 4.5$ cm, find EC .



Answer:

$$EC = 3\text{ cm}$$

Solution:

1. Since $DE \parallel BC$, by BPT, $AD/DB = AE/EC$.
2. So $3/2 = 4.5/EC$.
3. Hence $EC = 3$ cm.

Derivation:

$$\frac{3}{2} = \frac{4.5}{EC}$$
$$EC = 3$$

Triangles — basic proportionality theorem